

# Mark G. Scott

Chicopee, MA 01013 | +1 (732) 547-3217 | markgscott02@gmail.com | <https://markgscott960.github.io/Website/projects.html>

## SUMMARY

Data analyst and systems engineer with a background in quantitative social science research. Currently building and automating the operational data systems behind a home energy services contractor while completing an M.S. in Data Analytics and Computational Social Sciences. Comfortable across the full pipeline: research design, data collection and cleaning, modeling, and the systems that keep the data moving.

## TECHNICAL SKILLS

**Languages** R, Python

**Methods** Machine learning, network analysis, text analysis, spatial analysis, panel regression, research design

**Libraries** tidyverse, ggplot2, GeoPandas, linearmodels, PyTorch, OpenCV, btergm

**Tools** HubSpot, SPSS, Zapier, Stata, Git, Linux/WSL2

**Systems** Workflow automation, API integration, data pipeline design, regulatory compliance systems

## EDUCATION

### M.S., Data Analytics and Computational Social Sciences

Expected May 2027

*University of Massachusetts Amherst, Amherst, MA*

GPA 4.00/4.00

**Relevant coursework:** Machine Learning, Advanced Network Analysis, Advanced Text Analysis, Research Design and Methods, Spatial Analysis

### B.A., Political Science

May 2024

*Montclair State University, Montclair, NJ*

GPA 3.63/4.00

Dean's List 2023–2024 · Certificate of Recognition for Solving Authentic Complex Challenges (2024)

## EXPERIENCE

### Administrative Assitant → Systems Engineer

August 2025 – Present

*Green Collar, South Hadley, MA*

- Built an automated scheduling system for a sales team, replacing a manual assignment process.
- Designed a permitting workflow tracking municipal requirements jurisdiction by jurisdiction, and led the effort to bring the company into compliance with local jurisdictions' regulations and licensing requirements.
- Developed an estimation model to score and prioritize prospects, directing sales effort toward higher-value opportunities, and allowing for integration with business in other sectors.
- Administer the HubSpot CRM - workflow automation, data integrity, and integrations with external platforms.
- Led an SMS platform evaluation for the CRM: analyzed workflow data, shortlisted three vendors, and built a cost model that drove the selection decision.

### Research Assistant

January 2025 – August 2025

*APPRISE Inc., Princeton, NJ*

- Designed and implemented survey protocols for ongoing research projects.
- Developed methods that increased survey response rates, then standardized them across the research staff.
- Built and maintained a database tracking research activity across concurrent projects.

### Research Assistant

January 2024 – May 2024

*Montclair State University, Montclair, NJ*

- Collected, cleaned, and processed data from online databases, library archives, and expert interviews; produced literature summaries for project leads.
- Coded and labeled data from academic papers, surveys, and interviews for computational analysis.

## SELECTED PROJECTS

### Economic Dependence and UN Voting Alignment with China

2026

*Python · panel econometrics*

- Tested whether Chinese trade and investment dependence predicts UN voting alignment using panel fixed-effects regression (linearmodels PanelOLS) with cluster-robust standard errors.
- Adjudicated competing specifications with likelihood-ratio tests, AIC, and Wald tests, documenting why the cluster-robust result is authoritative.

## DOJ FOIA Corpus - Redaction Detection and Email Network Analysis

2025 – 2026

*Python, R · computer vision, network analysis*

- Built Python pipelines to scan roughly 1.38 million DOJ PDFs, extract email headers, and detect redactions with OpenCV and Mask R-CNN (Detectron2).
- Constructed a temporal email network in R: OCR name disambiguation via regex and `refinr`, then yearly network objects prepared for `btergm` modeling.